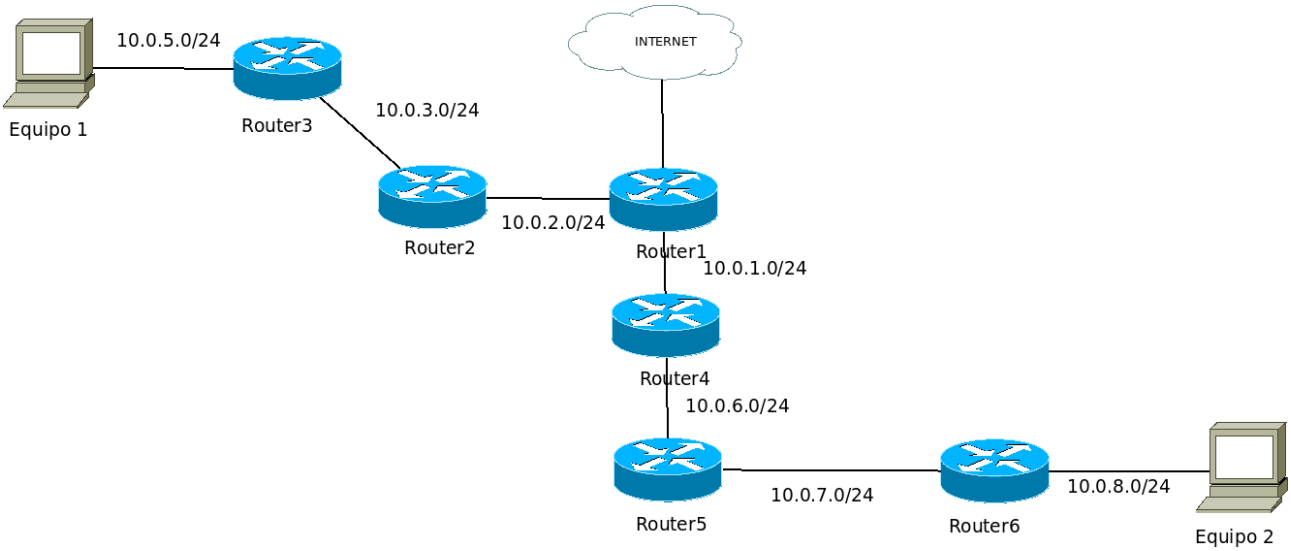
UT1. PRÁCTICA DE ENCAMINAMIENTO

1. Tenemos la siguiente red.



Genera el direccionamiento y las tablas de encaminamiento para comunicar todos los equipos y todos los routers.

Direccionamiento

Equipo	Interfaz	IP/red	Gateway	Servidor Nombres
Router1	**enp0s8 (internet)**	172.31.7.1/16	172.31.0.1/16	8.8.8.8
	enp0s9 (interna) R2	10.0.2.1/24		
	enp0s10 (interna)R4	10.0.1.1/24		
Router2	**enp0s9 (router1)**	10.0.2.2/24	10.0.2.1	8.8.8.8
	enp0s10(router3)	10.0.3.1/24		
Router3	**enp0s9 (router2)**	10.0.3.2/24	10.0.3.1	8.8.8.8
	enp0s10(equipo1)	10.0.5.1/24		
Router4	**enp0s9 (router1)**	10.0.1.2/24	10.0.1.1	8.8.8.8
	enp0s10(router5)	10.0.6.1/24		
Router5	**enp0s9 (router4)**	10.0.6.2/24	10.0.6.1	8.8.8.8
	enp0s10(router6)	10.0.7.1/24		
Router6	**enp0s9 (router5)**	10.0.7.2/24	10.0.7.1	8.8.8.8
	enp0s10(equipo2)	10.0.8.1/24		
Equipo1	**enp0s9 (router3)**	10.0.5.2/24	10.0.5.1	8.8.8.8
Equipo2	**enp0s9 (router6)**	10.0.8.8/24	10.0.8.1	8.8.8.8

Tabla de encaminamiento de los routers, añade las entradas que consideres necesarias.

Router	Destino	Gateway	Interfaz
Router1	10.0.3.0/24	10.0.2.2	enp0s9
	10.0.5.0/24	10.0.2.2	enp0s9
	10.0.6.0/24	10.0.1.2	enp0s10
	10.0.7.0/24	10.0.1.2	enp0s10
	10.0.8.0/24	10.0.1.2	enp0s10
Router2	10.0.5.0/24	10.0.3.2	enp0s10
Router4	10.0.7.0/24	10.0.6.2	enp0s10
	10.0.8.0/24	10.0.6.2	enp0s10
Router5	10.0.8.0/24	10.0.7.2	enp0s10

2. Simula el esquema de la red con equipos linux (sin interfaz gráfica). Genera el direccionamiento y el NAT para que todos los equipos tengan conexión entre sí y salgan a internet.

Solución:

Nota. Todas las redes pueden estar dentro de la misma red interna, menos la que sale a internet que es adaptador puente.

Router 1.

```
network:
    ethernets:
        enp0s3:
            dhcp4: true
        enp0s8:
            dhcp4: true
        enp0s9:
            dhcp4: false
            addresses:
              - 10.0.1.1/24
            nameservers:
              addresses:
                - 1.1.1.1
                - 8.8.8.8
            routes:
              - to: 10.0.6.0/24
                via: 10.0.1.2
              - to: 10.0.7.0/24
                via: 10.0.1.2
              - to: 10.0.8.0/24
                via: 10.0.1.2
        enp0s10:
            dhcp4: false
            addresses:
              - 10.0.2.1/24
            nameservers:
              addresses: [1.1.1.1, 8.8.8.8]
            routes:
              - to: 10.0.3.0/24
                via: 10.0.2.2
              - to: 10.0.5.0/24
                via: 10.0.2.2
    version: 2
```

Router 2.

```
network:
    ethernets:
        enp0s3:
            dhcp4: true
        enp0s8:
            dhcp4: false
            addresses:
              - 10.0.2.2/24
            nameservers:
              addresses: [1.1.1.1, 8.8.8.8]
            routes:
              - to: default
                via: 10.0.2.1
        enp0s9:
            dhcp4: false
            addresses:
              - 10.0.3.1/24
            nameservers:
              addresses: [1.1.1.1, 8.8.8.8]
            routes:
              - to: 10.0.5.0/24
                via: 10.0.3.2
        enp0s10:
            dhcp4: true
    version: 2
```

Router 3.

```
network:
    ethernets:
        enp0s3:
            dhcp4: true
        enp0s8:
            dhcp4: false
            addresses:
              - 10.0.3.2/24
            nameservers:
              addresses:
                - 1.1.1.1
                - 8.8.8.8
            routes:
              - to: default
                via: 10.0.3.1
        enp0s9:
            dhcp4: false
            addresses:
              - 10.0.5.1/24
        enp0s10:
            dhcp4: true
    version: 2
```

Router 4.

```
network:
    ethernets:
        enp0s3:
            dhcp4: true
        enp0s8:
            dhcp4: false
            addresses:
              - 10.0.1.2/24
            nameservers:
              addresses:
                - 1.1.1.1
                - 8.8.8.8
            routes:
              - to: default
                via: 10.0.1.1
        enp0s9:
            dhcp4: false
            addresses:
              - 10.0.6.1/24
            routes:
              - to: 10.0.7.0/24
                via: 10.0.6.2
              - to: 10.0.8.0/24
                via: 10.0.6.2
        enp0s10:
            dhcp4: true
    version: 2
```

Router 5.

```
network:
    ethernets:
        enp0s3:
            dhcp4: true
        enp0s8:
            dhcp4: false
            addresses:
              - 10.0.6.2/24
            nameservers:
              addresses:
                - 1.1.1.1
                - 8.8.8.8
            routes:
              - to: default
                via: 10.0.6.1
        enp0s9:
            dhcp4: false
            addresses:
              - 10.0.7.1/24
            nameservers:
              addresses:
                - 1.1.1.1
                - 8.8.8.8
            routes:
              - to: 10.0.8.0/24
                via: 10.0.7.2
        enp0s10:
            dhcp4: true
    version: 2
```

Router 6.

```
network:
    ethernets:
        enp0s3:
            dhcp4: true
        enp0s8:
            dhcp4: false
            addresses:
              - 10.0.7.2/24
            nameservers:
              addresses:
                - 1.1.1.1
                - 8.8.8.8
            routes:
              - to: default
                via: 10.0.7.1
        enp0s9:
            dhcp4: false
            addresses:
              - 10.0.8.1/24
        enp0s10:
            dhcp4: true
    version: 2
```

Conseguir NAT.

Después de configurar todo para que todos los routers se vean entre si y se comuniquen todos los equipos lo que hay que conseguir es que todas las redes salgan a internet a través de NAT.

```
alumno@Router1:~$ sudo iptables -t nat -L -nv
Chain PREROUTING (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

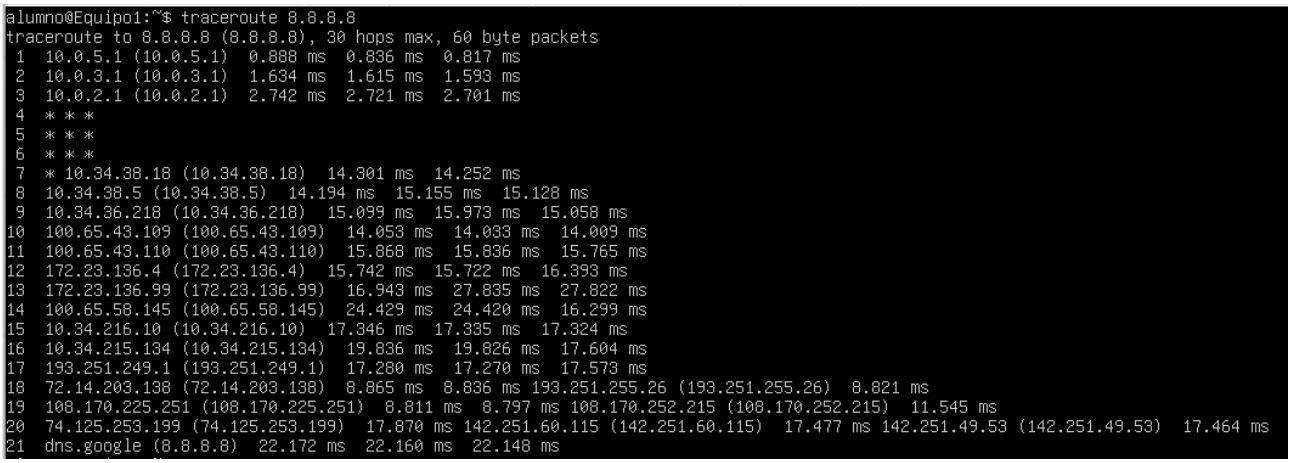
Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain POSTROUTING (policy ACCEPT 93 packets, 11915 bytes)
 pkts bytes target     prot opt in     out     source               destination
   12   898 MASQUERADE  0    --  *      enp0s8  10.0.1.0/24          0.0.0.0/0
   15  1126 MASQUERADE  0    --  *      enp0s8  10.0.2.0/24          0.0.0.0/0
   10   736 MASQUERADE  0    --  *      enp0s8  10.0.3.0/24          0.0.0.0/0
    9   680 MASQUERADE  0    --  *      enp0s8  10.0.5.0/24          0.0.0.0/0
   20  1466 MASQUERADE  0    --  *      enp0s8  10.0.6.0/24          0.0.0.0/0
    3   216 MASQUERADE  0    --  *      enp0s8  10.0.7.0/24          0.0.0.0/0
    1    64 MASQUERADE  0    --  *      enp0s8  10.0.8.0/24          0.0.0.0/0
```

```
alumno@Router1:~$ sudo cat  /etc/iptables/rules.v4
# Generated by iptables-save v1.8.10 (nf_tables) on Tue Sep 29 10:54:22 2026
*nat
:PREROUTING ACCEPT [0:0]
:INPUT ACCEPT [0:0]
:OUTPUT ACCEPT [0:0]
:POSTROUTING ACCEPT [93:11915]
-A POSTROUTING -s 10.0.1.0/24 -o enp0s8 -j MASQUERADE
-A POSTROUTING -s 10.0.2.0/24 -o enp0s8 -j MASQUERADE
-A POSTROUTING -s 10.0.3.0/24 -o enp0s8 -j MASQUERADE
-A POSTROUTING -s 10.0.5.0/24 -o enp0s8 -j MASQUERADE
-A POSTROUTING -s 10.0.6.0/24 -o enp0s8 -j MASQUERADE
-A POSTROUTING -s 10.0.7.0/24 -o enp0s8 -j MASQUERADE
-A POSTROUTING -s 10.0.8.0/24 -o enp0s8 -j MASQUERADE
COMMIT
# Completed on Tue Sep 29 10:54:22 2026
```

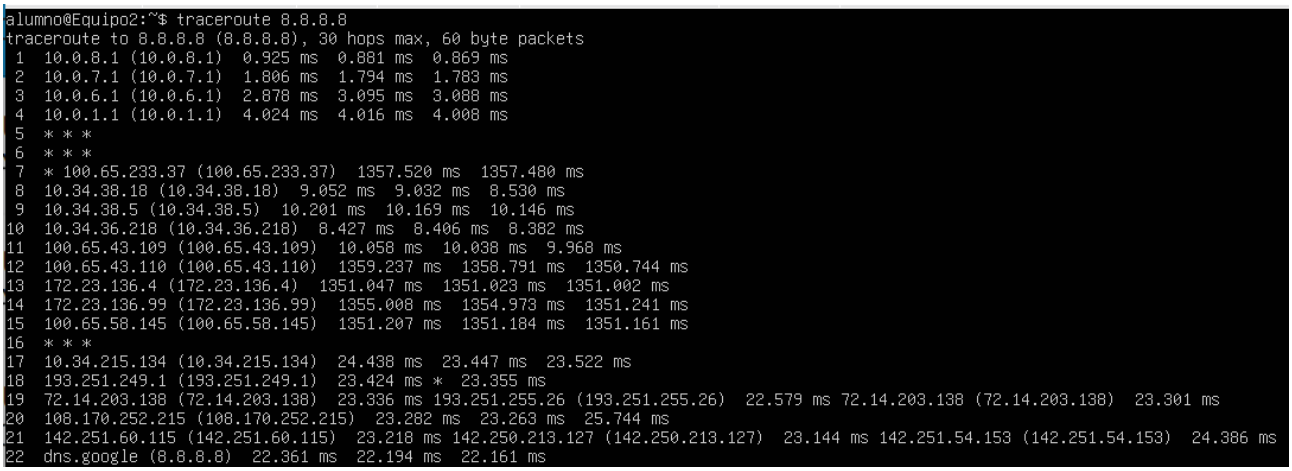
Conseguir Traceroute.

Equipo1 a internet:



```
alumno@Equipo1:~$ traceroute 8.8.8.8
traceroute to 8.8.8.8 (8.8.8.8), 30 hops max, 60 byte packets
 1  10.0.5.1 (10.0.5.1)  0.888 ms  0.836 ms  0.817 ms
 2  10.0.3.1 (10.0.3.1)  1.634 ms  1.615 ms  1.593 ms
 3  10.0.2.1 (10.0.2.1)  2.742 ms  2.721 ms  2.701 ms
 4  * * *
 5  * * *
 6  * * *
 7  * 10.34.38.18 (10.34.38.18)  14.301 ms  14.252 ms
 8  10.34.38.5 (10.34.38.5)  14.194 ms  15.155 ms  15.128 ms
 9  10.34.36.218 (10.34.36.218)  15.099 ms  15.973 ms  15.058 ms
10  100.65.43.109 (100.65.43.109)  14.053 ms  14.033 ms  14.009 ms
11  100.65.43.110 (100.65.43.110)  15.868 ms  15.836 ms  15.765 ms
12  172.23.136.4 (172.23.136.4)  15.742 ms  15.722 ms  16.393 ms
13  172.23.136.99 (172.23.136.99)  16.943 ms  27.835 ms  27.822 ms
14  100.65.58.145 (100.65.58.145)  24.429 ms  24.420 ms  16.299 ms
15  10.34.216.10 (10.34.216.10)  17.346 ms  17.335 ms  17.324 ms
16  10.34.215.134 (10.34.215.134)  19.836 ms  19.826 ms  17.604 ms
17  193.251.249.1 (193.251.249.1)  17.280 ms  17.270 ms  17.573 ms
18  72.14.203.138 (72.14.203.138)  8.865 ms  8.836 ms 193.251.255.26 (193.251.255.26)  8.821 ms
19  108.170.225.251 (108.170.225.251)  8.811 ms  8.797 ms 108.170.252.215 (108.170.252.215)  11.545 ms
20  74.125.253.199 (74.125.253.199)  17.870 ms 142.251.60.115 (142.251.60.115)  17.477 ms 142.251.49.53 (142.251.49.53)  17.464 ms
21  dns.google (8.8.8.8)  22.172 ms  22.160 ms  22.148 ms
```

Equipo2 a internet:



```
alumno@Equipo2:~$ traceroute 8.8.8.8
traceroute to 8.8.8.8 (8.8.8.8), 30 hops max, 60 byte packets
 1  10.0.8.1 (10.0.8.1)  0.925 ms  0.881 ms  0.869 ms
 2  10.0.7.1 (10.0.7.1)  1.806 ms  1.794 ms  1.783 ms
 3  10.0.6.1 (10.0.6.1)  2.878 ms  3.095 ms  3.088 ms
 4  10.0.1.1 (10.0.1.1)  4.024 ms  4.016 ms  4.008 ms
 5  * * *
 6  * * *
 7  * 100.65.233.37 (100.65.233.37)  1357.520 ms  1357.480 ms
 8  10.34.38.18 (10.34.38.18)  9.052 ms  9.032 ms  8.530 ms
 9  10.34.38.5 (10.34.38.5)  10.201 ms  10.169 ms  10.146 ms
10  10.34.36.218 (10.34.36.218)  8.427 ms  8.406 ms  8.382 ms
11  100.65.43.109 (100.65.43.109)  10.058 ms  10.038 ms  9.968 ms
12  100.65.43.110 (100.65.43.110)  1359.237 ms  1358.791 ms  1350.744 ms
13  172.23.136.4 (172.23.136.4)  1351.047 ms  1351.023 ms  1351.002 ms
14  172.23.136.99 (172.23.136.99)  1355.008 ms  1354.973 ms  1351.241 ms
15  100.65.58.145 (100.65.58.145)  1351.207 ms  1351.184 ms  1351.161 ms
16  * * *
17  10.34.215.134 (10.34.215.134)  24.438 ms  23.447 ms  23.522 ms
18  193.251.249.1 (193.251.249.1)  23.424 ms *  23.355 ms
19  72.14.203.138 (72.14.203.138)  23.336 ms 193.251.255.26 (193.251.255.26)  22.579 ms 72.14.203.138 (72.14.203.138)  23.301 ms
20  108.170.252.215 (108.170.252.215)  23.282 ms  23.263 ms  25.744 ms
21  142.251.60.115 (142.251.60.115)  23.218 ms 142.250.213.127 (142.250.213.127)  23.144 ms 142.251.54.153 (142.251.54.153)  24.386 ms
22  dns.google (8.8.8.8)  22.361 ms  22.194 ms  22.161 ms
```